# Akoizo Original Practice - 2027 Rotation Form 3

**Division C | Circuit Lab | 50-minute written practice**

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

**Name: _____ Date: _____**

**1. (2 points) An LED’s forward voltage is…**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Answer: _____

**2. (2 points) A watt-hour measures…**

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Answer: _____

**3. (2 points) Negative solved current means…**

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. No voltage exists

B. Opposite the assumed direction

C. Impossible circuit necessarily

D. Negative charge disappears

Answer: _____

**4. (2 points) A diode is generally…**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. A mechanical switch

B. Nonlinear

C. An ideal fixed resistor

D. Always conducting both ways

Answer: _____

**5. (4 points) Why is charge not consumed?**

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

**6. (4 points) Write a source-two-resistor loop equation.**

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

**7. (4 points) Name two LED faults.**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

**8. (4 points) Why is Ah not sufficient for battery energy?**

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

Answer: _____
_____
_____

**9. (4 points) Explain a nodal branch current.**

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

**10. (4 points) Why not infer brightness from current alone?**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

**11. (2 points) A 9 V ideal battery drives a 4.5 ohm resistor. Find current.**

Answer: _____
_____
_____

**12. (2 points) A series pair has resistances 5 and 7 ohms. Find equivalent resistance.**

Answer: _____
_____
_____

**13. (2 points) Two identical 10 ohm resistors are placed in parallel. Find equivalent resistance.**

Answer: _____
_____
_____

**14. (2 points) An ideal resistor carries 2 A at 9 V. Find power.**

Answer: _____
_____
_____

**15. (2 points) A node has incoming currents 6 A and 2 A and one outgoing current. Find the outgoing current.**

Answer: _____
_____
_____

**16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.**

Answer: _____
_____
_____

**17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.**

Answer: _____
_____
_____

**18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.**

Answer: _____
_____
_____

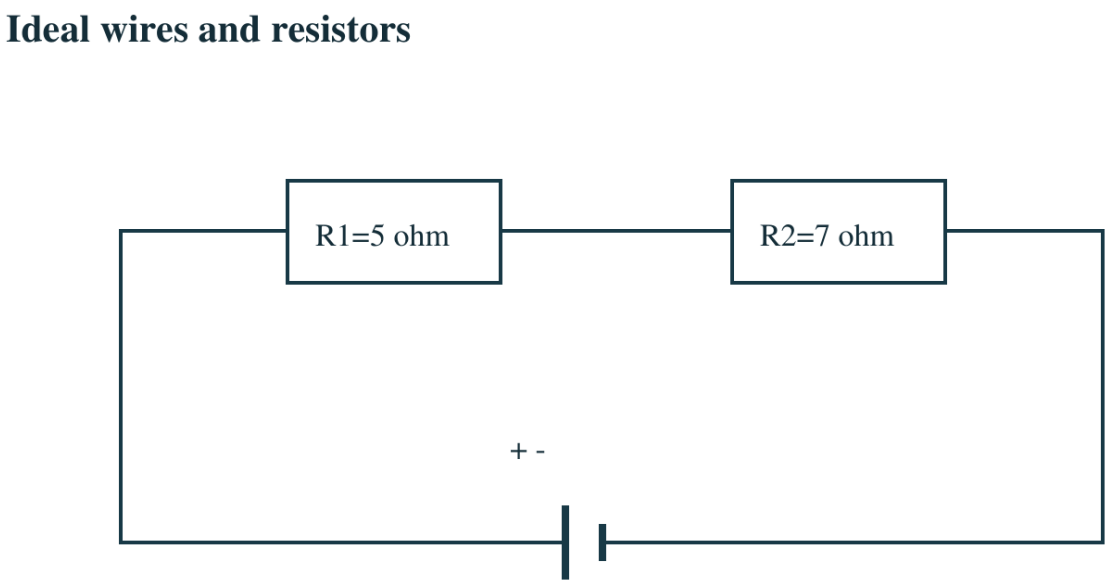
**19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.**

Answer: _____
_____
_____

**20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.**

Answer: _____
_____
_____

**21. (2 points) Use Figure F1. Calculate equivalent resistance.**




F1. Series resistor circuit (original synthetic teaching figure)

Answer: _____
_____
_____

**22. (2 points) Use Figure F1. Calculate loop current.**

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

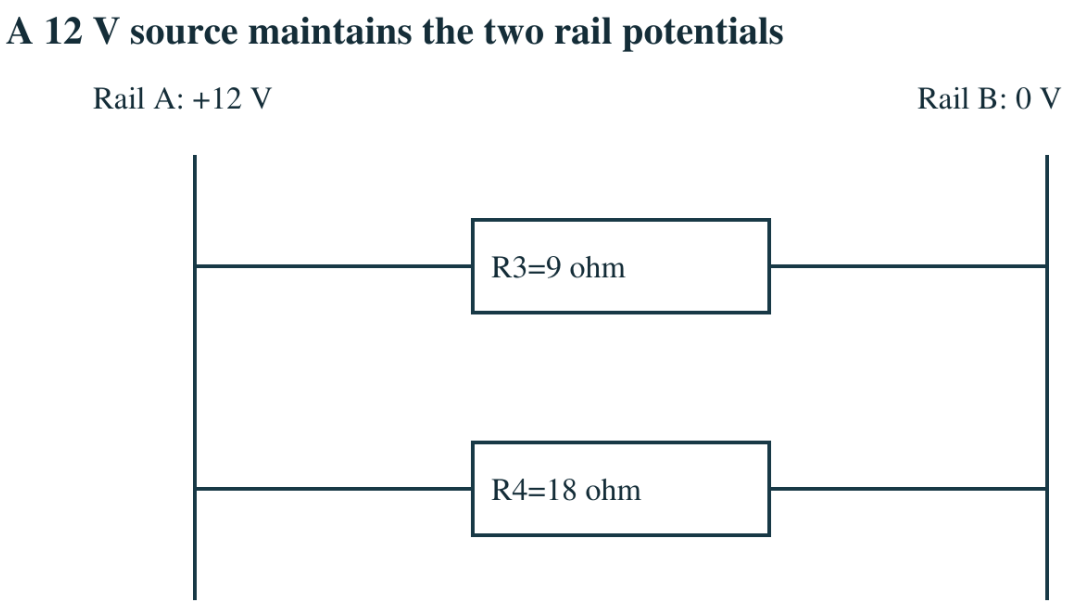
Answer: _____
_____
_____

**23. (2 points) Use Figure F1. Calculate the potential difference across R1.**

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**24. (2 points) Use Figure F2. Find current through R3.**




F2. Two parallel branches (original synthetic teaching figure)

Answer: _____
_____
_____

**25. (2 points) Use Figure F2. Find current through R4.**

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

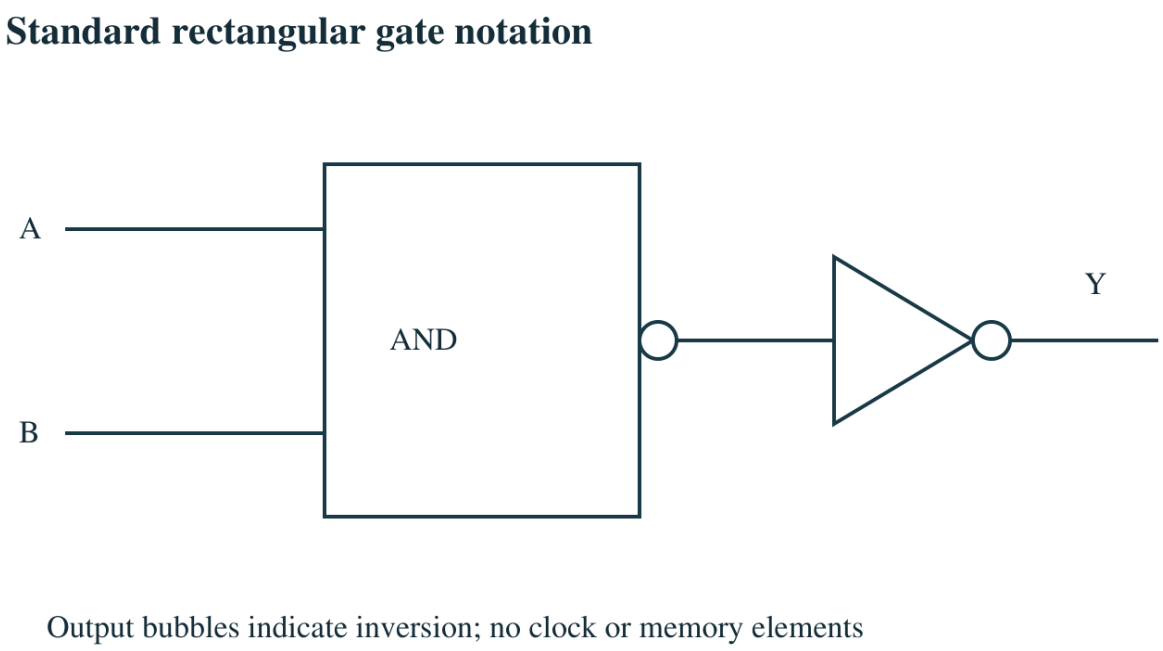
Answer: _____
_____
_____

**26. (2 points) Use Figure F2. Find equivalent resistance between the rails.**

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**27. (2 points) Use Figure F3. Find Y for A=1 and B=0.**




F3. Combinational logic (original synthetic teaching figure)

Answer: _____
_____
_____

**28. (2 points) Use Figure F3. Find Y for A=1 and B=1.**

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

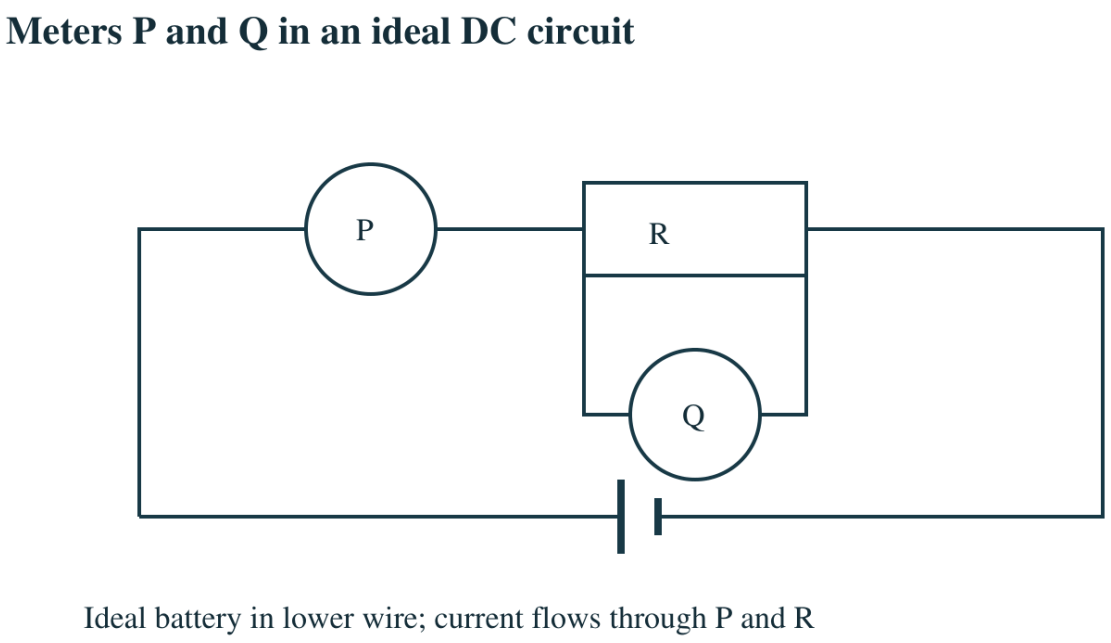
Answer: _____
_____
_____

**29. (2 points) Use Figure F3. Name the single gate equivalent to the complete diagram and justify.**

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**30. (2 points) Use Figure F4. Which meter position is appropriate for a voltmeter? Explain using the wiring.**




F4. Measurement connections (original synthetic teaching figure)

Answer: _____
_____
_____

**31. (2 points) Use Figure F4. Which meter position is appropriate for an ammeter? Explain.**

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**32. (2 points) Use Figure F4. Describe the desired input resistance of the meter at Q and its purpose.**

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____